The Pericardium & Heart



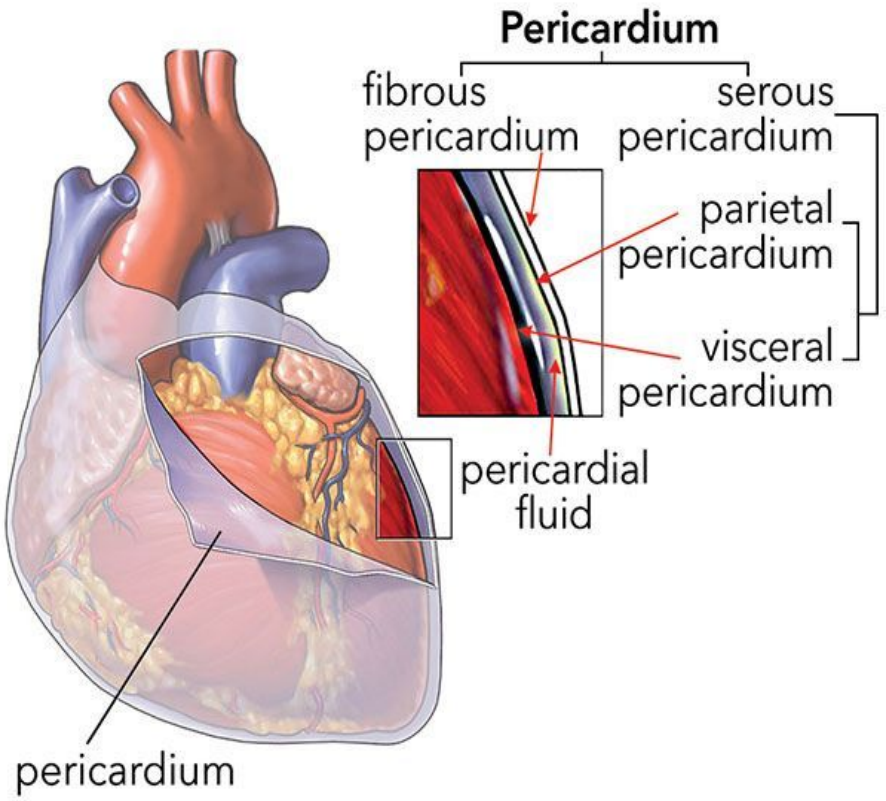
Learning Objectives:

At the end of the lecture, the students should be able to:

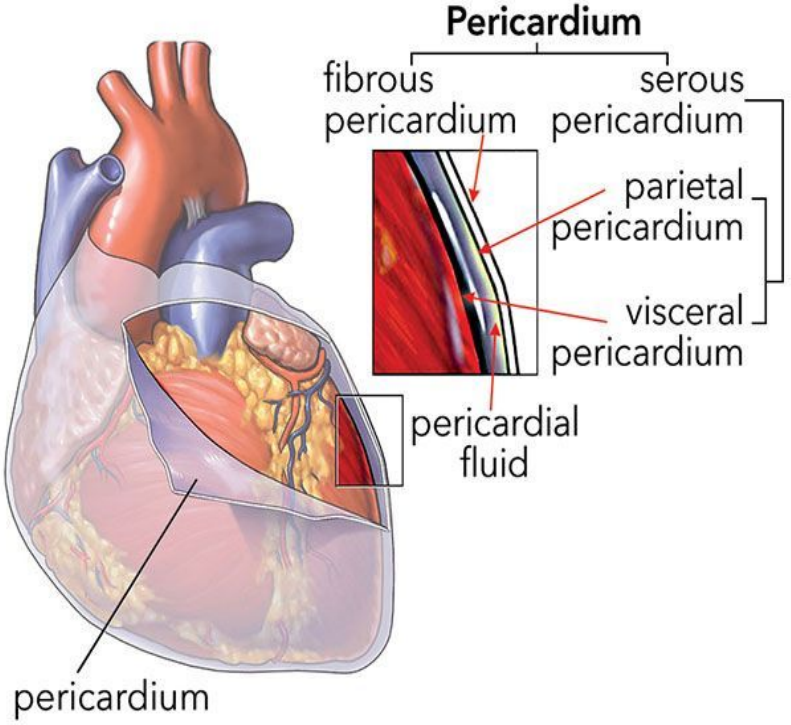
- **Define pericardium, its layers, & its nerve supply.**
- **Explain routes of circulation through the heart.**
- **Describe surfaces & borders of the heart.**
- **Enumerate chambers of human heart.**
- **Describe main external & internal features of each chamber and their partitioning.**
- **Define nerve supply of heart.**

The Pericardium:

- It is a fibroserous sac which encloses the heart and roots of great blood vessels.
- It further consists of two layers:
- **The Fibrous Pericardium:** it is made up of fibrous tissue and encloses the heart and fuses with the roots of great blood vessels that enter or exit the heart.

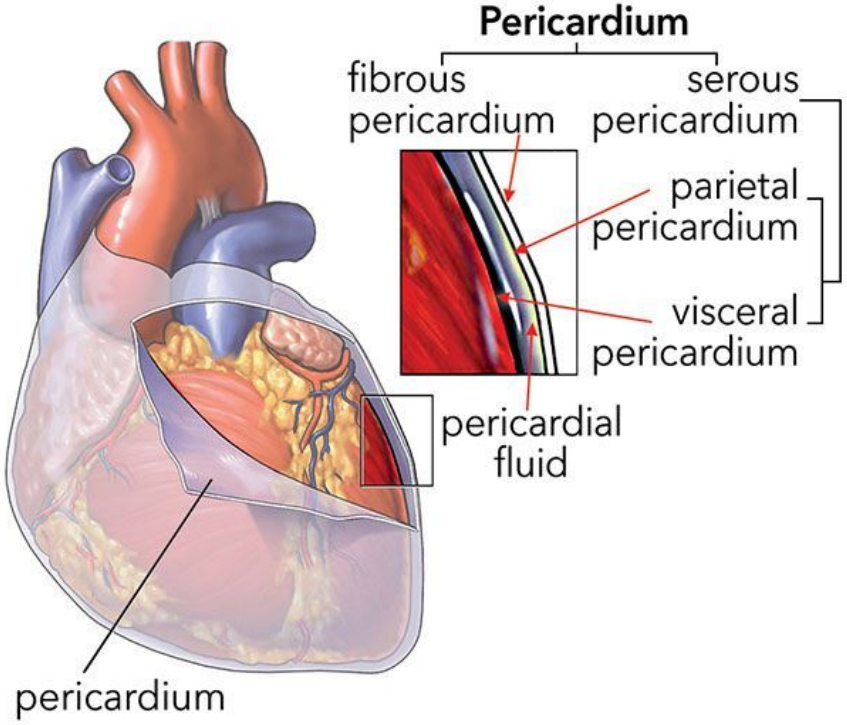


- **The Serous Pericardium:** it is a double-layered membrane lined by mesothelium.
- The outer layer/parietal pericardium is fused with fibrous pericardium,
- Whereas the inner visceral layer/epicardium is fused to the heart except at cardiac grooves where is separated from heart by blood vessels.
- The two layers are continuous at roots of great blood vessels i.e. ascending aorta, pulmonary trunk, two venae cavae and four pulmonary veins.



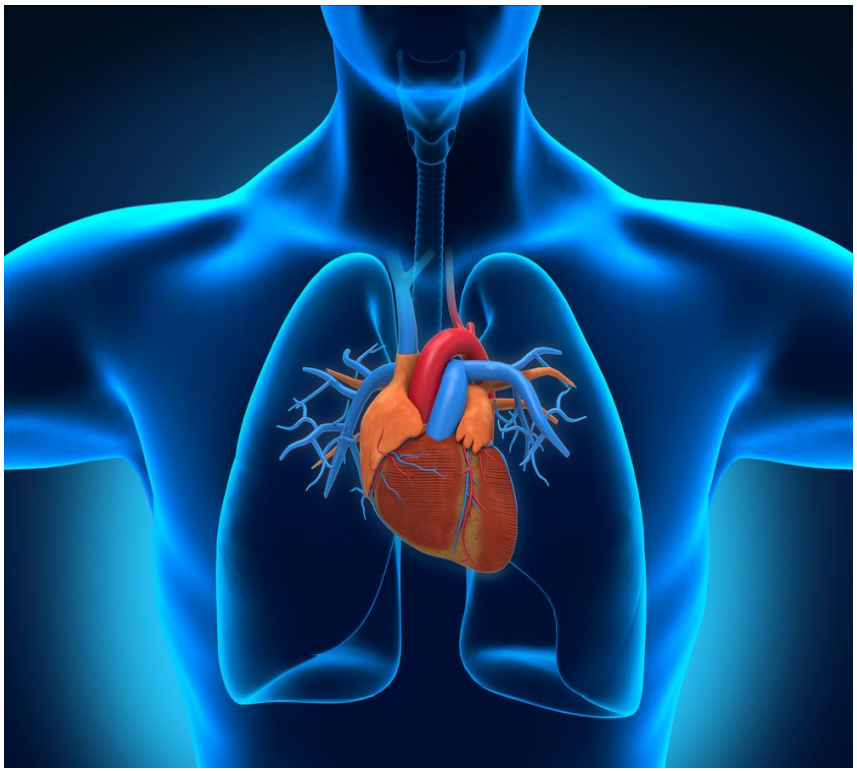
Pericardial Cavity:

- The potential space between parietal and visceral pericardium/epicardium is called pericardial cavity which contains thin, serous fluid, the pericardial fluid.



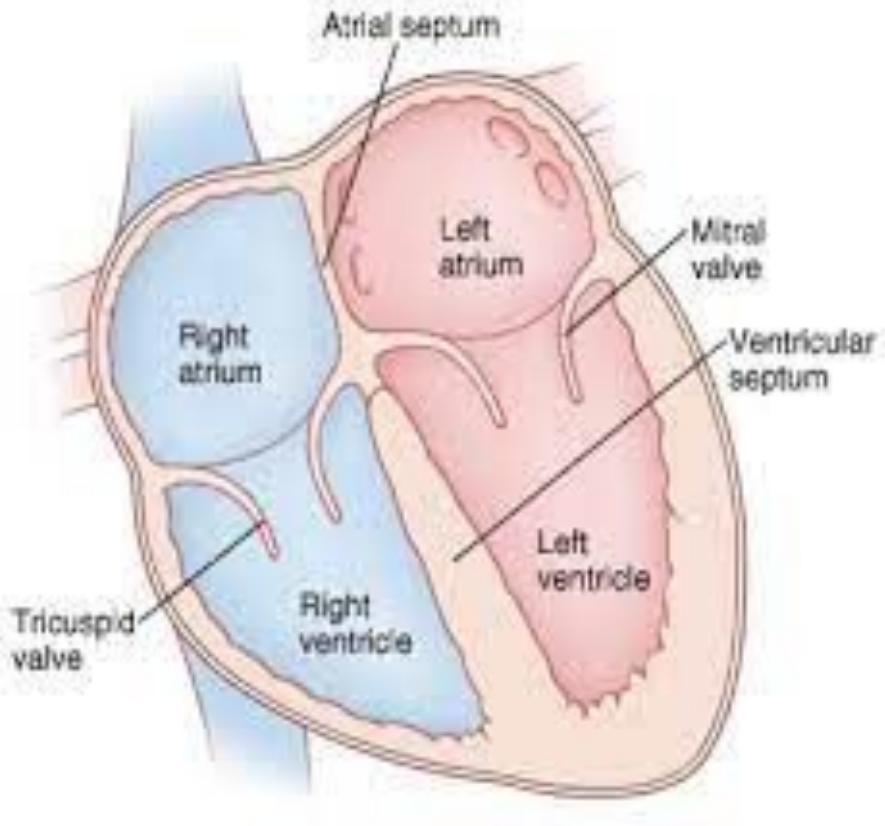
The Heart - Function & Location:

- It is a hollow muscular organ.
- It is situated in the middle mediastinum.
- It pumps blood to different parts of the body to meet their nutritive requirements.
- It is placed obliquely behind the sternum and the adjoining parts of costal cartilages, so that 1/3rd of it lies to the right and 2/3rd of it to the left of median plane.



Chambers of the Heart:

- The human heart has 4 chambers.
- These are the **right & left ATRIA** and the **right & left VENTRICLES.**
- The right atrium receives deoxygenated blood from the whole body & the right ventricle sends it to the lungs.
- Whereas, the left atrium receives oxygenated blood from the lungs & the left ventricle sends it to the body.

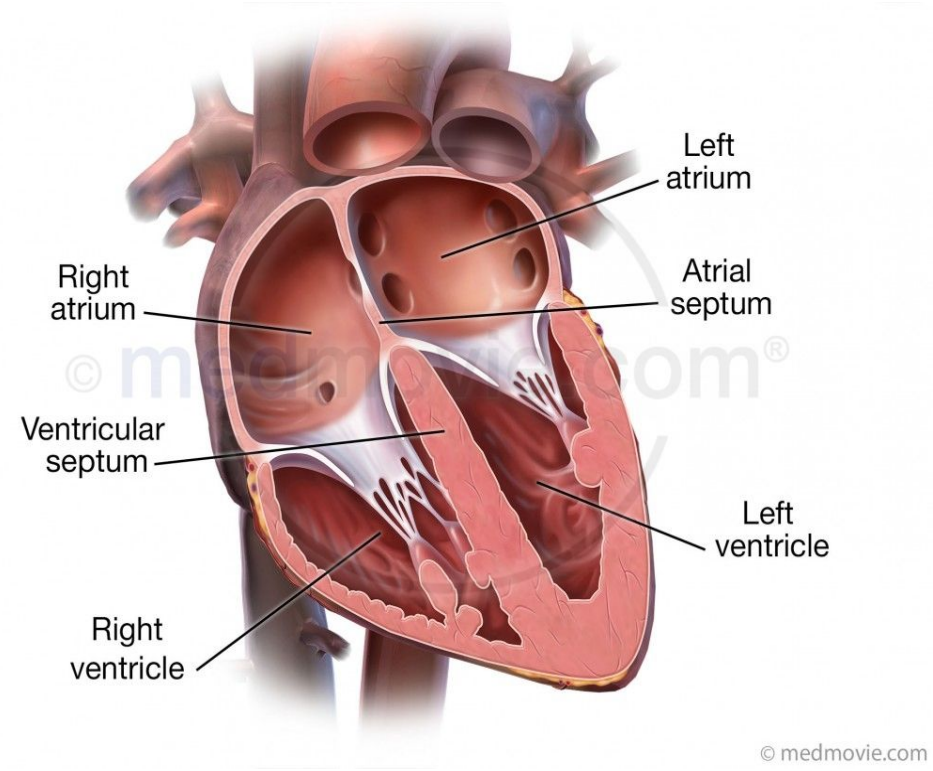


Nerve Supply of Pericardium:

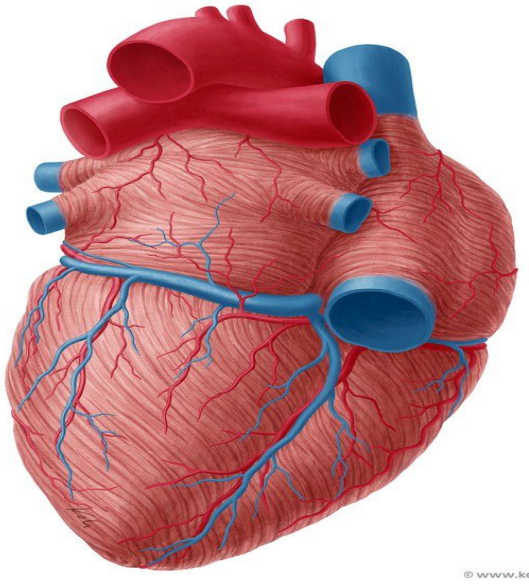
- The ***fibrous & parietal pericardia*** are supplied by **Phrenic nerves.**
- They are sensitive to pain.
- The ***visceral pericardium (epicardium)*** is supplied by **autonomic nerves of the heart**, and is not sensitive to pain.

External Features of the Heart:

- The **two atria** are separated from each other by **inter-atrial septum** and the **two ventricles** are partitioned by **inter-ventricular septum.**
- The **atria** lie **above** the **ventricles** and are separated from ventricles by **atrio-ventricular groove (coronary sulcus).**

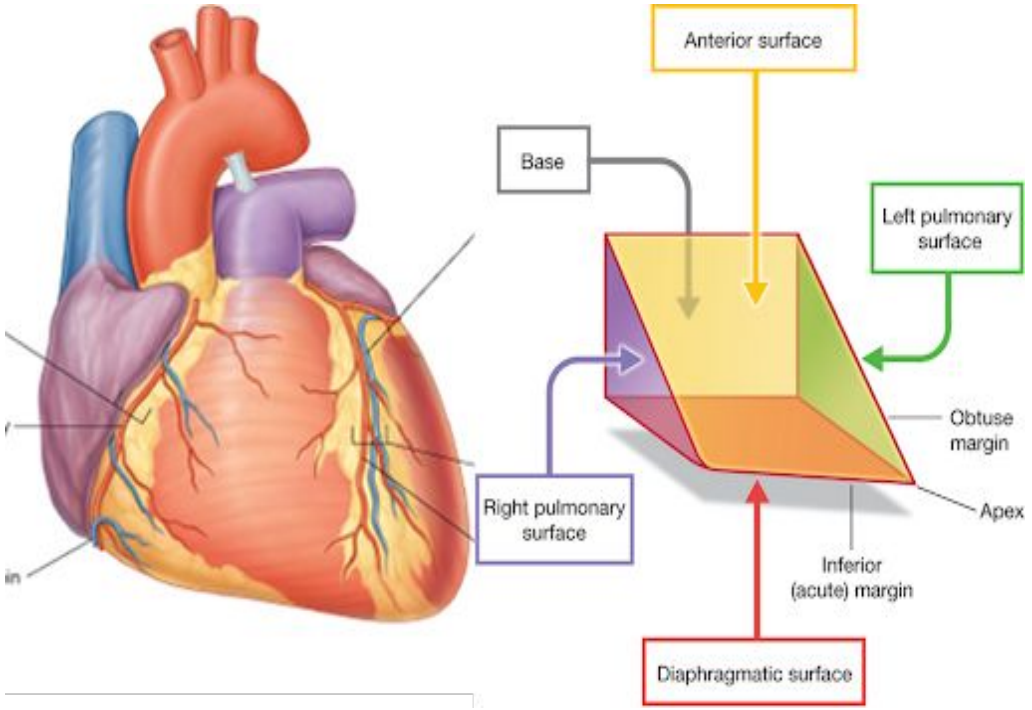


Surfaces of the Heart:



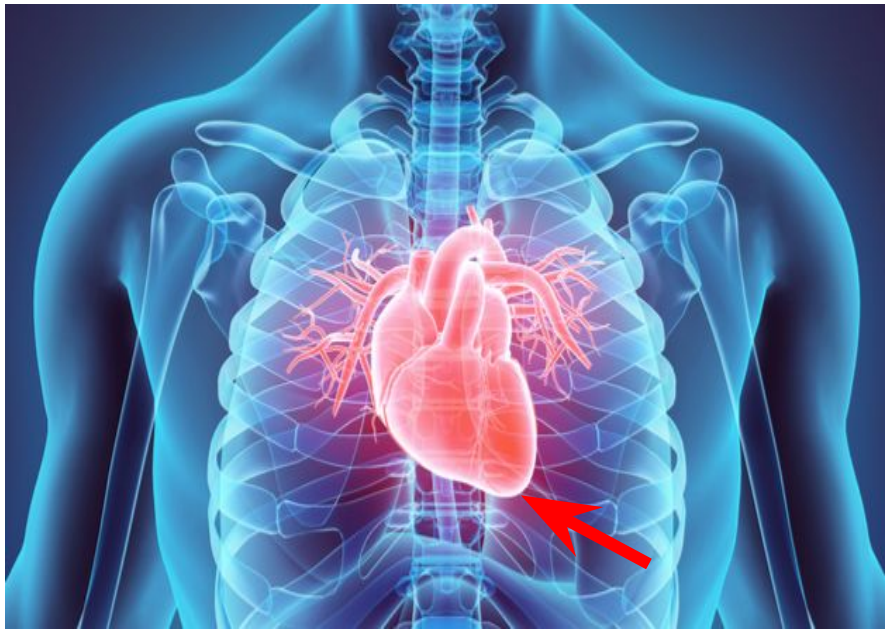
The heart has:

- An **Apex**, directed downwards towards the left side,
- A **Base/posterior surface** directed backwards,
- An **Anterior/ sternocostal surface,**
- An **Inferior surface,** &
- A **Left lateral surface.**



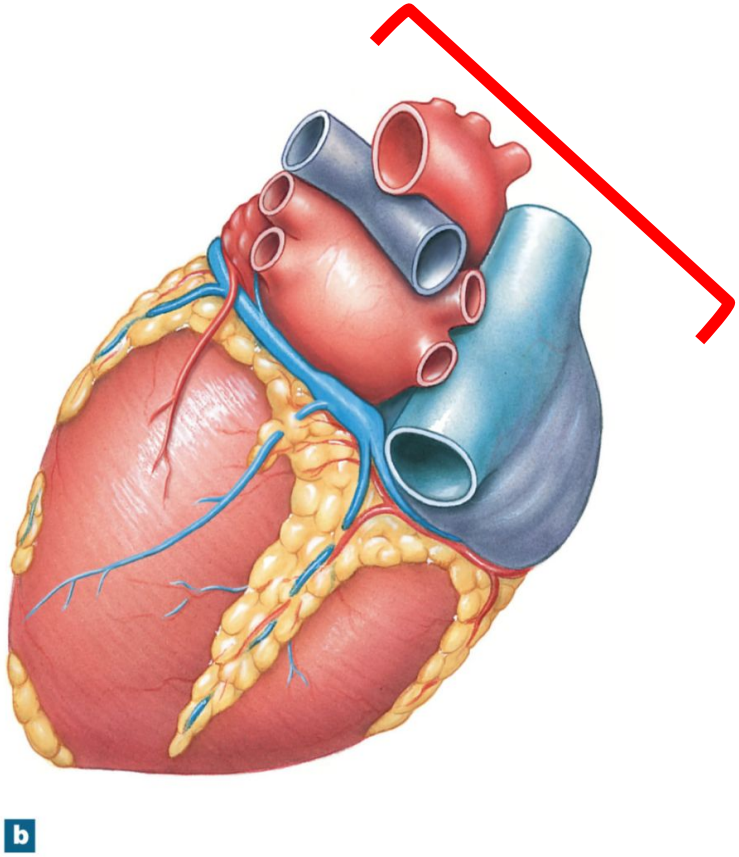
Apex of the Heart:

- The **Apex** is formed entirely by ***left ventricle*.**
- It is overlapped by anterior border of left lung.
- It is situated in the ***left 5th intercostal space*** about 9 cm lateral to mid-sternal line.
- Pulsations can be seen/felt over this part.



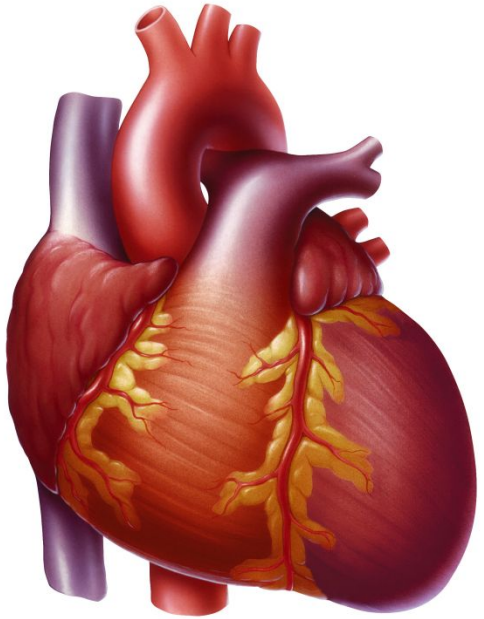
Base of the Heart:

- The **Base/posterior surface** is mainly formed by ***left atrium*** and by a small part of right atrium.
- The openings that can be seen in relation to posterior surface are:
- Openings of 4 pulmonary veins into the left atrium, &
- Openings of superior & inferior venae cavae into the right atrium.



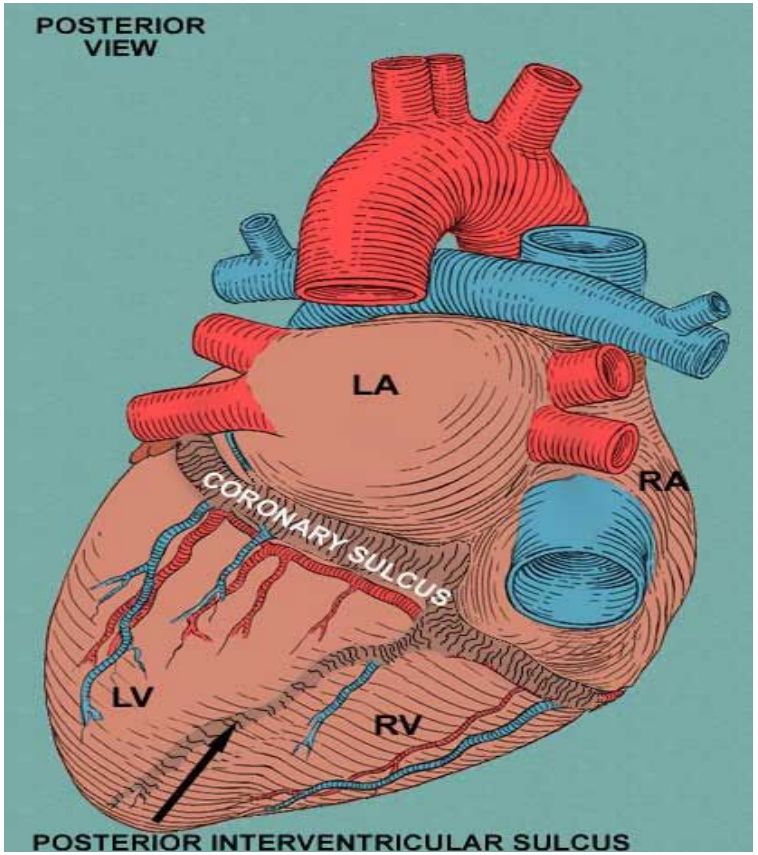
The Sternocostal surface:

- The **Anterior/sternocostal surface** is formed mainly by ***Right Atrium & Right Ventricle*** and partly by Left Atrium & Left Atrium.
- Most of the anterior surface is covered by lungs, but a part of it that lies behind cardiac notch remains uncovered, called *Superficial Area of Cardiac Dullness.*



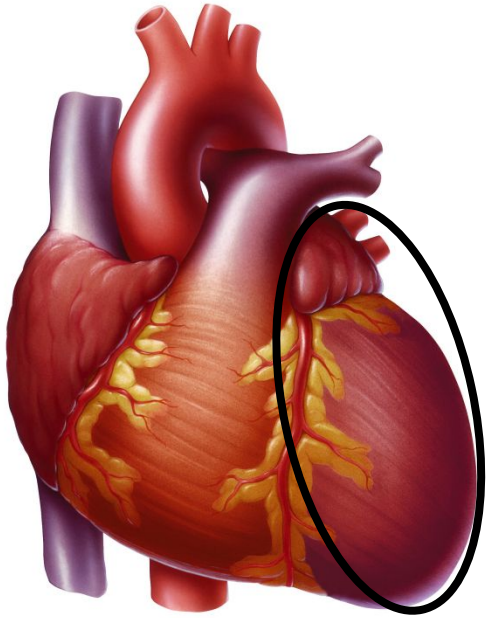
The Diaphragmatic surface:

- The Inferior/diaphragmatic surface rests on the central tendon of the diaphragm.
- This surface is formed in its left 2/3rd by left ventricle & by right ventricle in its right 1/3rd.
- It is crossed by posterior part of inter-ventricular groove.



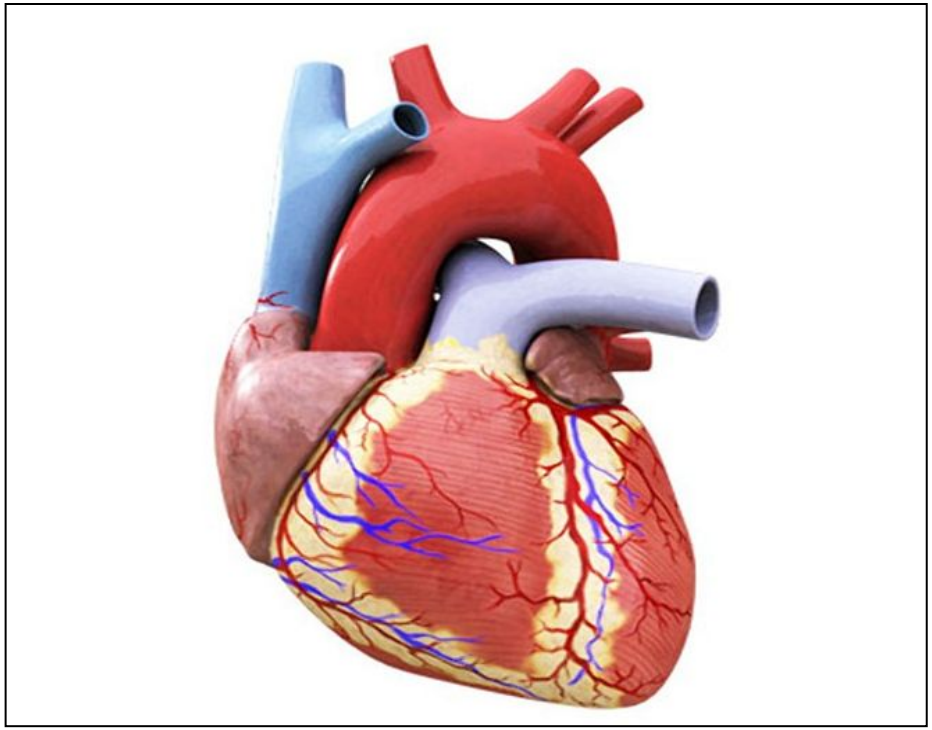
The Left surface:

- The Left surface is mostly formed by left ventricle & by left auricle at its upper end.



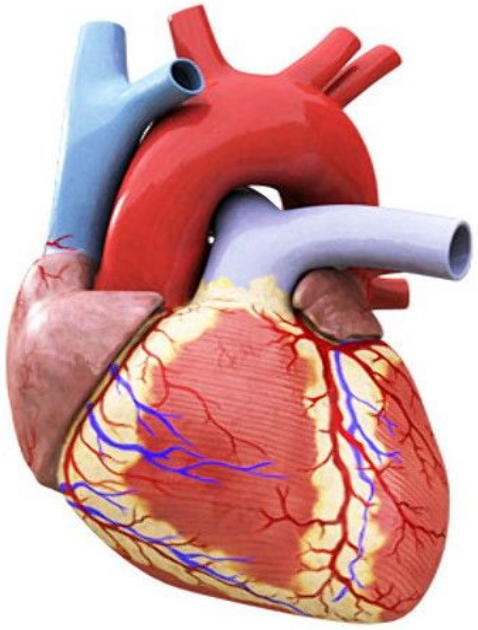
Borders of the Heart:

- The surfaces of the heart are demarcated by **Upper, Inferior, Right & Left Borders.**
- The **Upper border** is oblique and formed by Two Atria, mainly the ***left atrium.***
- The **Right border** is vertical and formed by ***Right Atrium***. It extends from superior vena cava to inferior vena cava.
- The **Inferior border** is horizontal and mainly formed by ***Right Ventricle*** & partly by left ventricle. It extends from superior vena cava to the apex.
- The **Left border** is oblique. It is formed mainly by ***Left Ventricle*** & partly by left auricle.



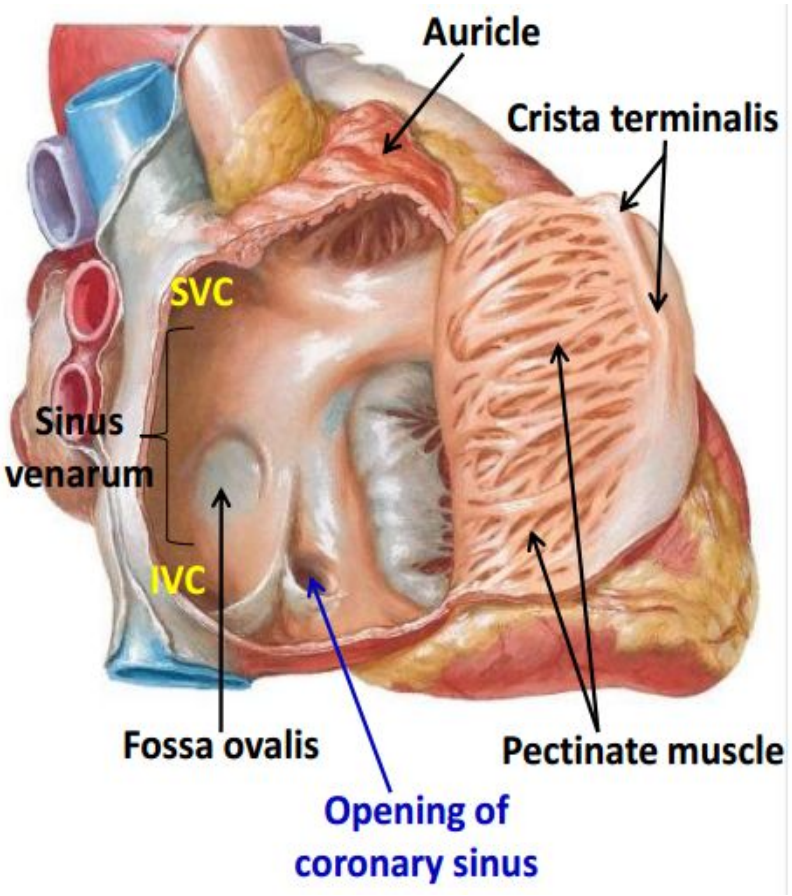
The Right Atrium:

- It is the upper right chamber of the heart.
- It receives venous blood from the whole body and pumps it to the right ventricle via right atrioventricular/tricuspid opening.
- Its upper end is prolonged to the left to form right auricle.
- The interior of the auricle is sponge-like.



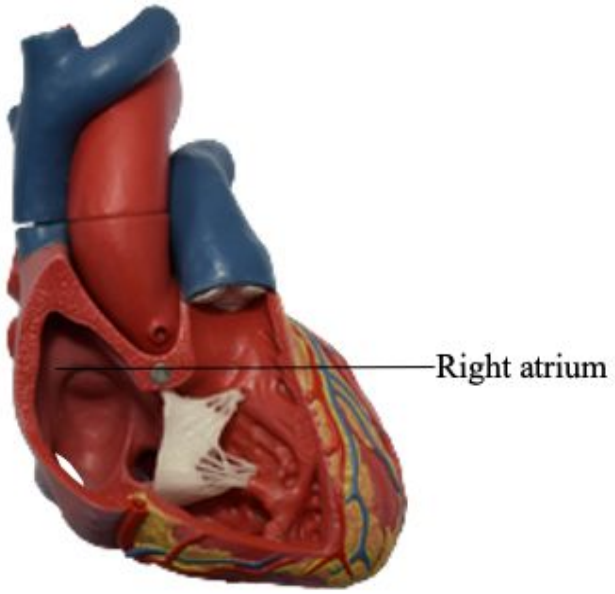
Features of the Right Atrium:

- **Sulcus terminalis:** it is a vertical groove which passes from superior vena cava above to the inferior vena cava below.
- This groove is produced by an internal muscular ridge called **crista terminalis.**
- The upper part of sulcus/groove contains sinoatrial **(SA) node or pacemaker** of the heart.



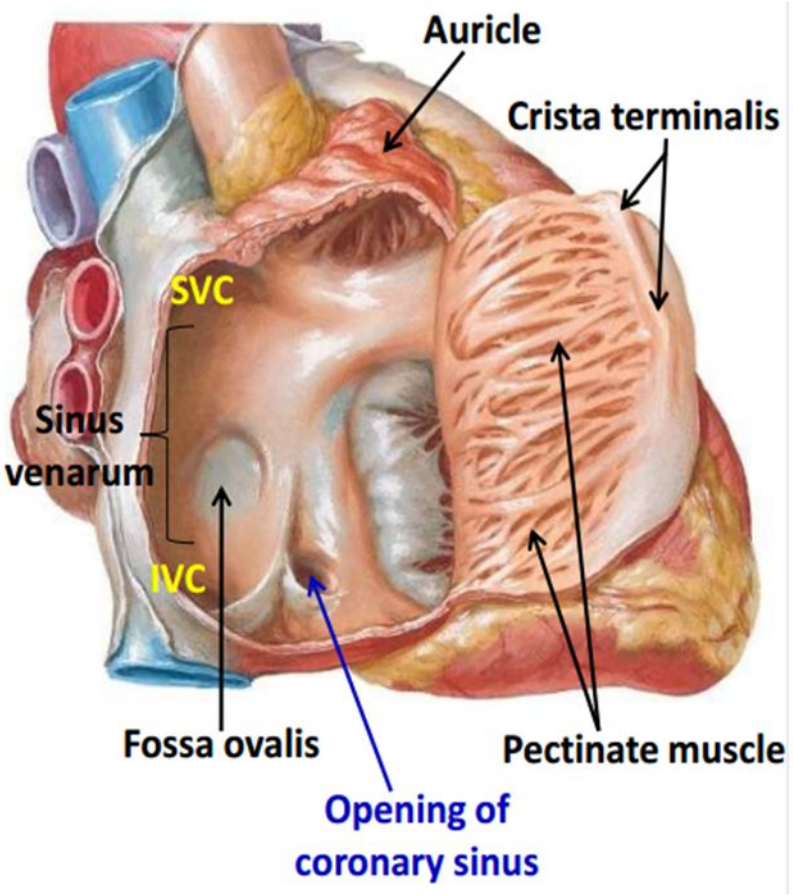
The Right Atrioventricular Orifice:

- The right atrium sends blood to the right ventricle through atrioventricular/tricuspid orifice.
- The above orifice is guarded by tricuspid valve to maintain unidirectional blood flow.



Tributaries/orifices of Right Atrium:

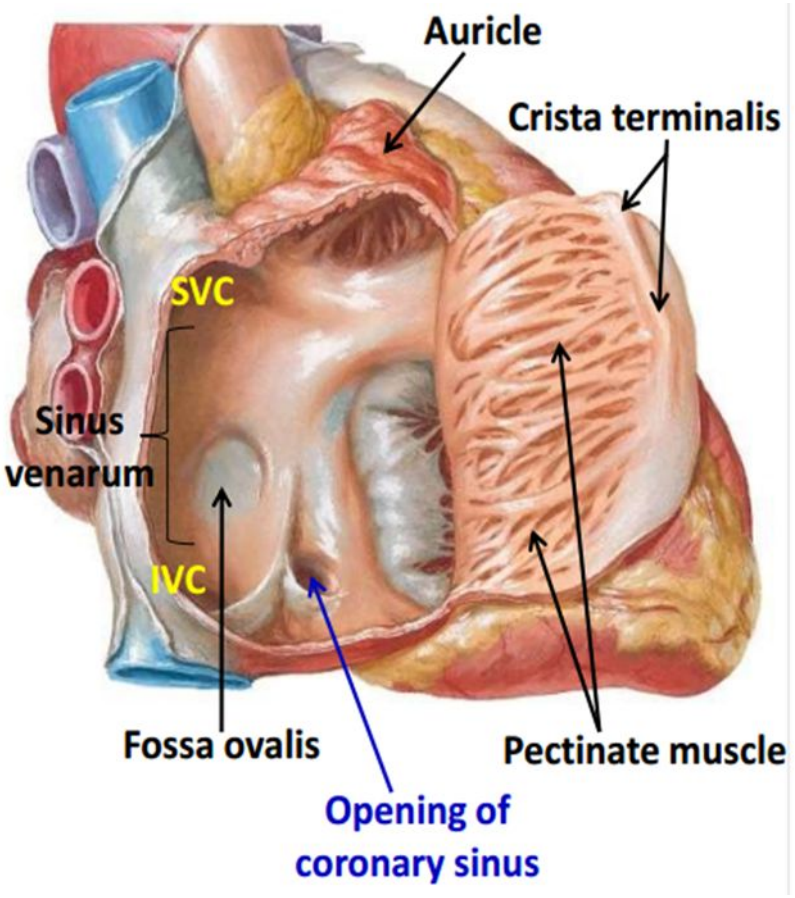
- **Superior vena cava:** at the upper end,
- **Inferior vena cava:** at the lower end & guarded by eustachian valve,
- **Coronary sinus:** between the opening of inferior vena cava & right atrioventricular orifice. It is guarded by thebesian valve,
- **Venae cordis minimae/thebesian veins:** these small veins are present in all 4 chambers of the heart and open in right atrium through small foramina.



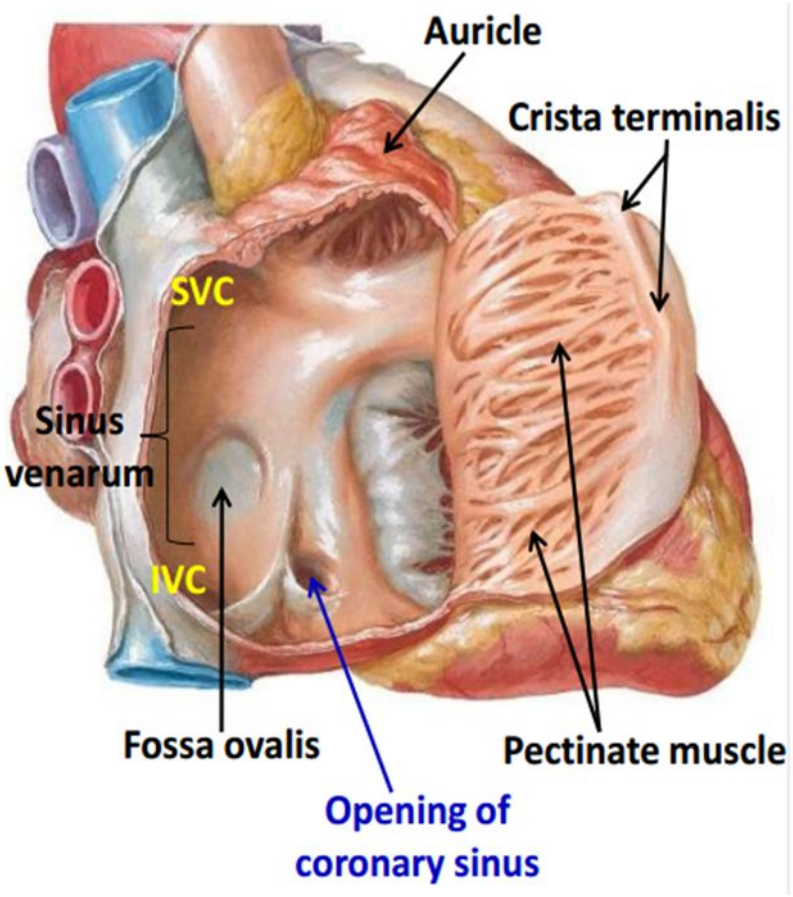
Interior of Right Atrium:

- Internally, the right atrium is divided into three parts:

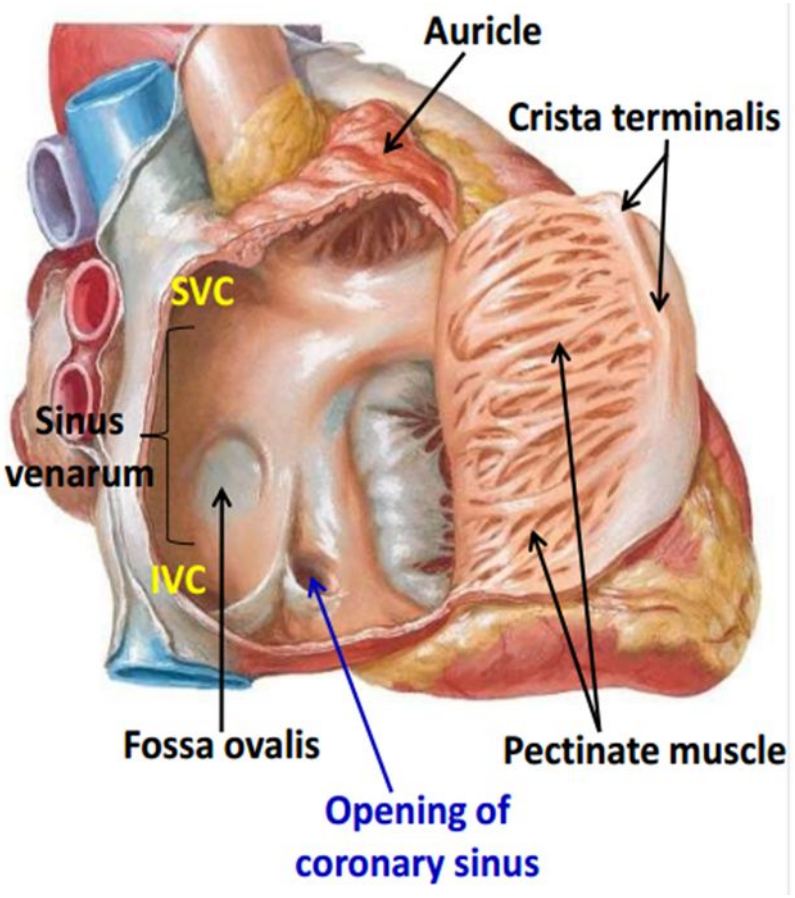
i. **Smooth posterior part (sinus venarum):** it is developed from right horn of sinus venosus and it receives all the tributaries except for anterior cardiac veins.



ii. **Rough anterior part (pectinate part):** it is developed from primitive atrial chamber and it contains transverse muscular ridges called **musculi pectinati.** These ridges arise from crista terminalis and end in atrioventricular orifice.



iii. **Interatrial septum:** it is developed from septum primum & septum secundum. It contains **fossa ovalis** (the embryonic remnant of septum primum. The **limbus/annulus ovalis** is the prominent margin of fossa ovalis but it is deficient inferiorly.



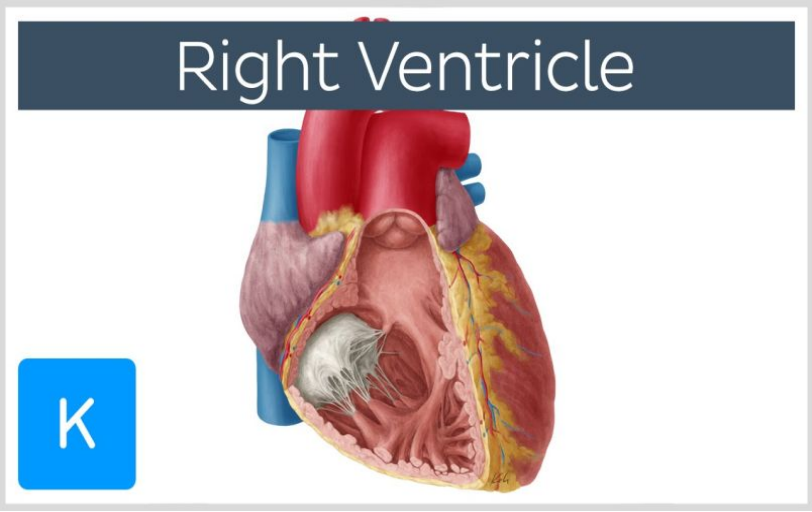
The Right Ventricle:

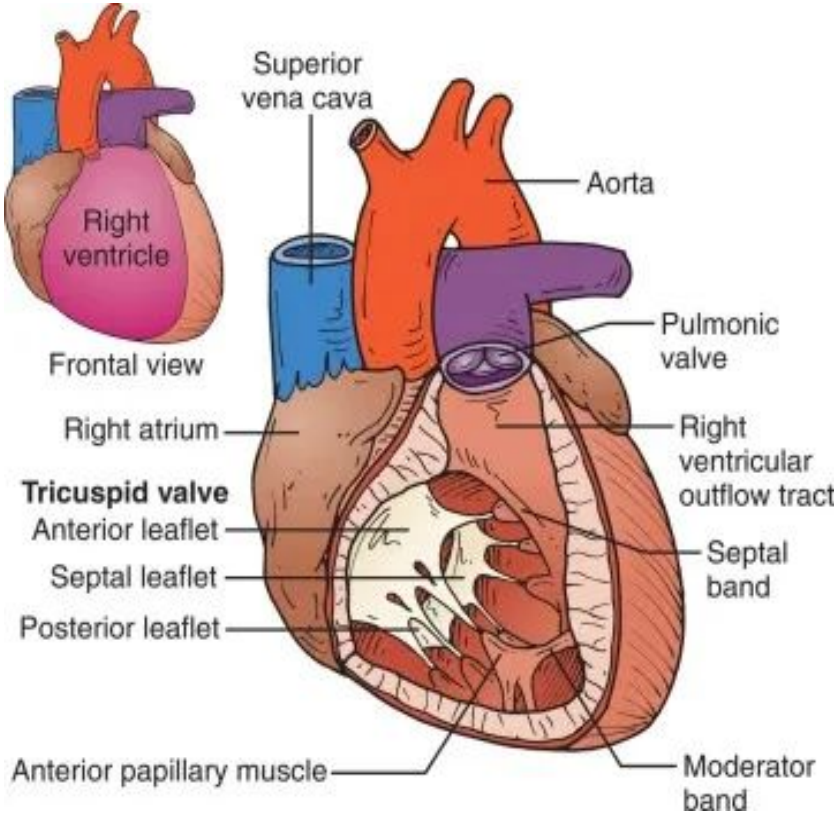
- It is a triangular chamber which receives blood from right atrium and pumps it to the lungs by pulmonary arteries.
- Internally, right ventricle has two parts:

i. **The rough Inflowing part** due to the presence of **muscular ridges/trabeculae carnae**. The muscular ridges may be*: ridges, bridges* or *pillars (papillary muscles).*

✔ There are 3 papillary muscles in right ventricle: anterior, posterior & septal.

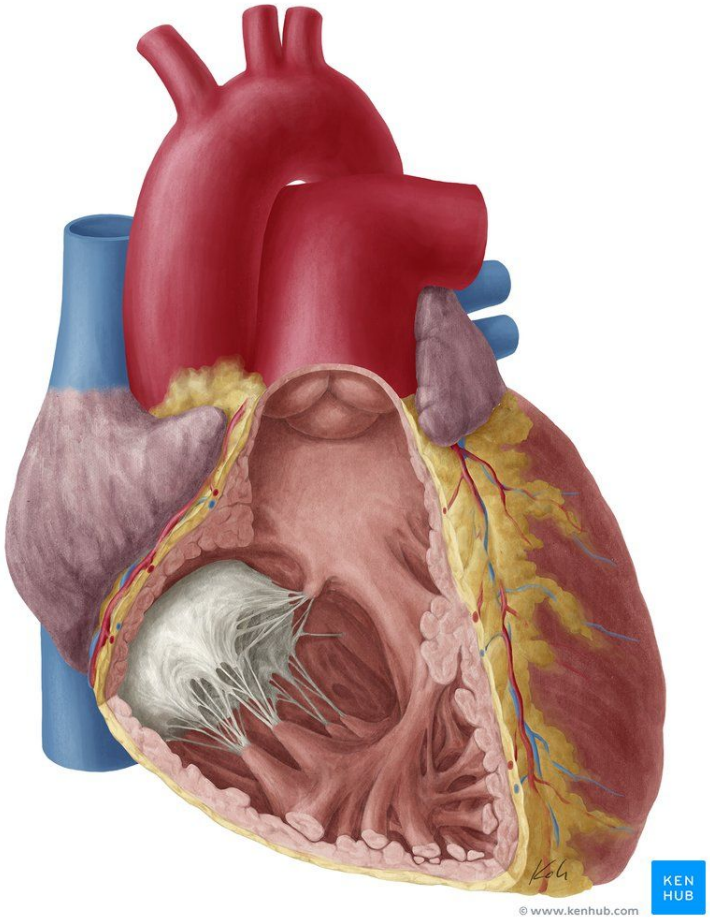
- **Septomarginal trabecula/moderator band:** it is a muscular ridge extending from ventricular septum to anterior papillary muscle and contains right branch of atrioventricular (AV) bundle of His.





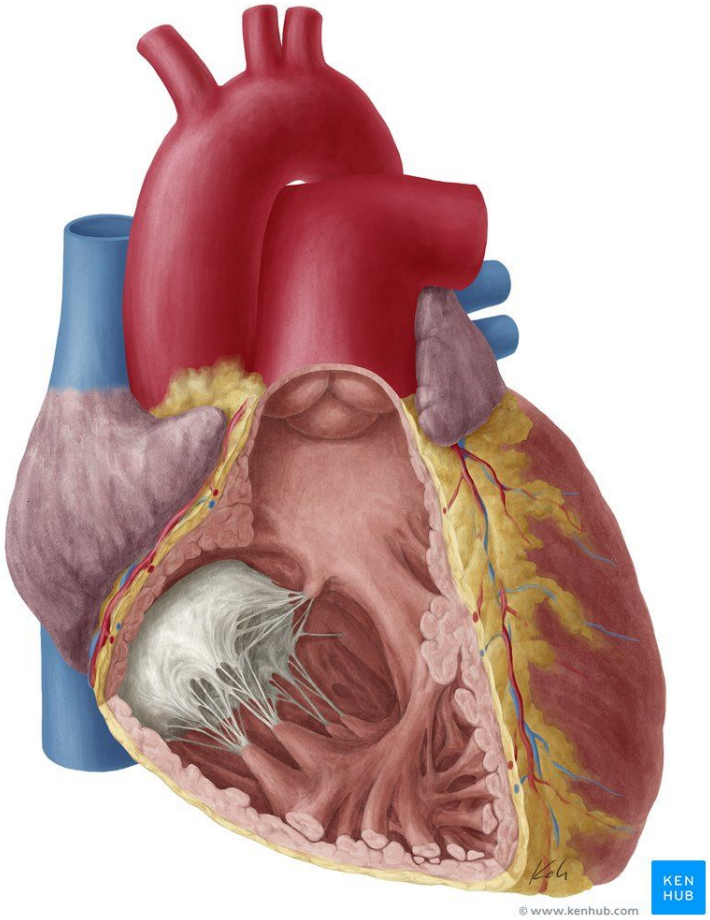
ii. **The smooth Outflowing part (infundinbulum):** it forms upper conical part. It develops from bulbus cordis and gives origin to pulmonary trunk.

- The outflow and inflow tracts are separated by a muscular ridge, the *supraventricular crest* that lies between tricuspid and pulmonary openings.



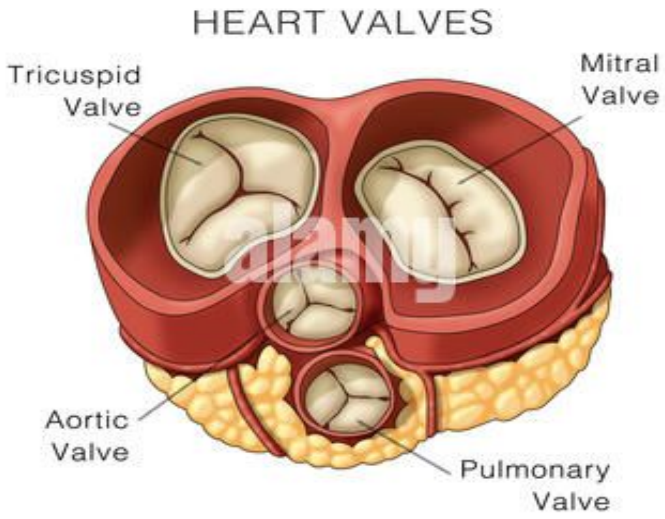
The Tricuspid & Pulmonary Orifices:

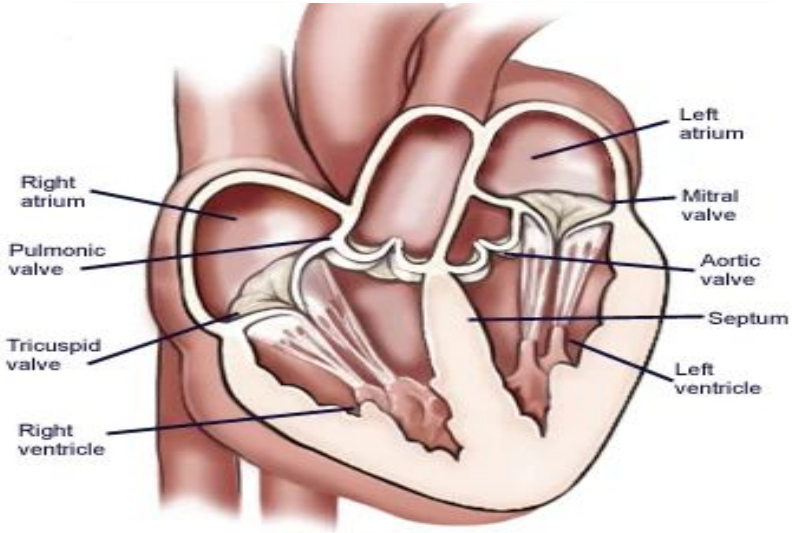
- The tricuspid/right atrioventricular opening guarded by tricuspid valve, &
- The pulmonary orifice guarded by semilunar (pulmonary) valve.
- The valves of the heart maintain unidirectional flow of blood and prevent its regurgitation in opposite direction.



The Right & Left Atrioventricular (Tricuspid & Bicuspid) Valves:

- There are **two pairs of valves** in the heart, a **pair of atrioventricular valves,** and a **pair of semilunar valves.**
- The **semilunar valves** include ***aortic & pulmonary valves,*** each having three cusps.
- The **right atrioventricular valve** is known as ***tricuspid valve*** because it has three cusps, whereas **left atrioventricular valve** is known as ***bicuspid valve*** because it has two cusps.
- The cusps are folds of endocardium, reinforced by intervening fibrous tissue.





Structure of Atrioventricular Valves:

- Both atrioventricular valves are made up of the following:

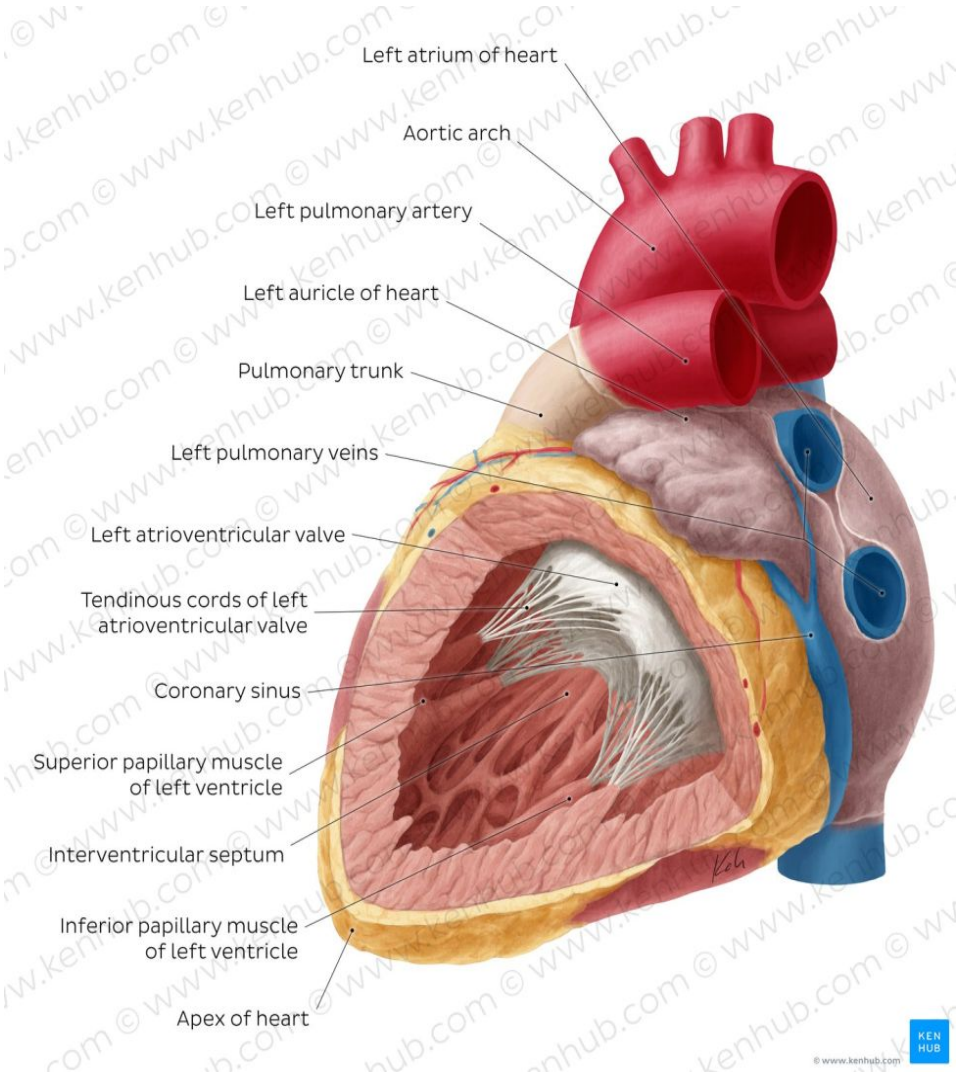
a) A fibrous ring to which the cusps are attached.

b) Each cusp has an attached and a free margin.

c) The free margins are attached to chordae tendinae.

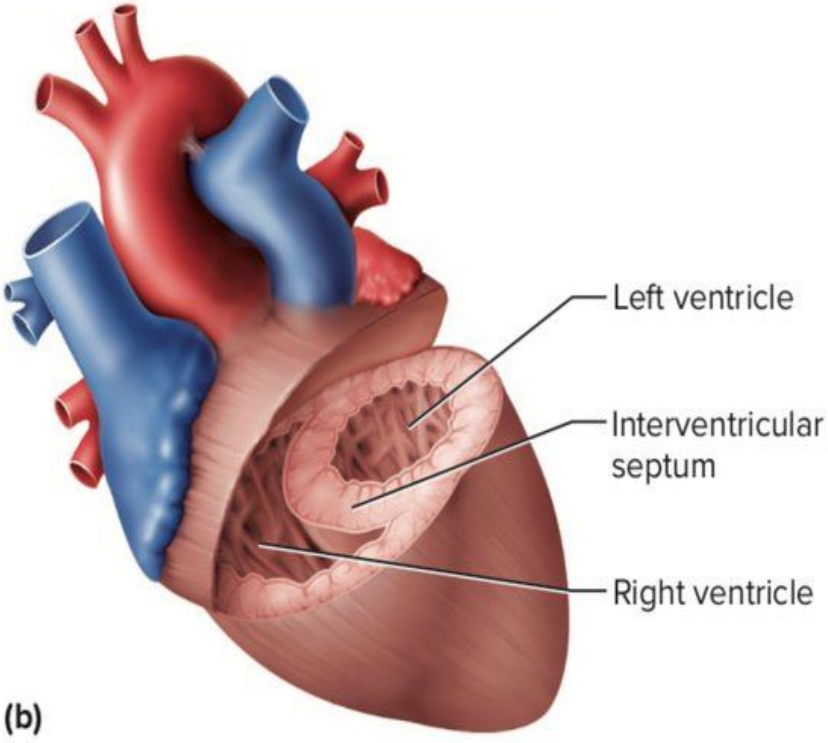
d) The chordae tendinae connect the free margins and ventricular surfaces of cusps to papillary muscles.

e) The atrioventricular valves are kept competent by active contraction of papillary muscles which prevent the eversion of free margins of cusps.



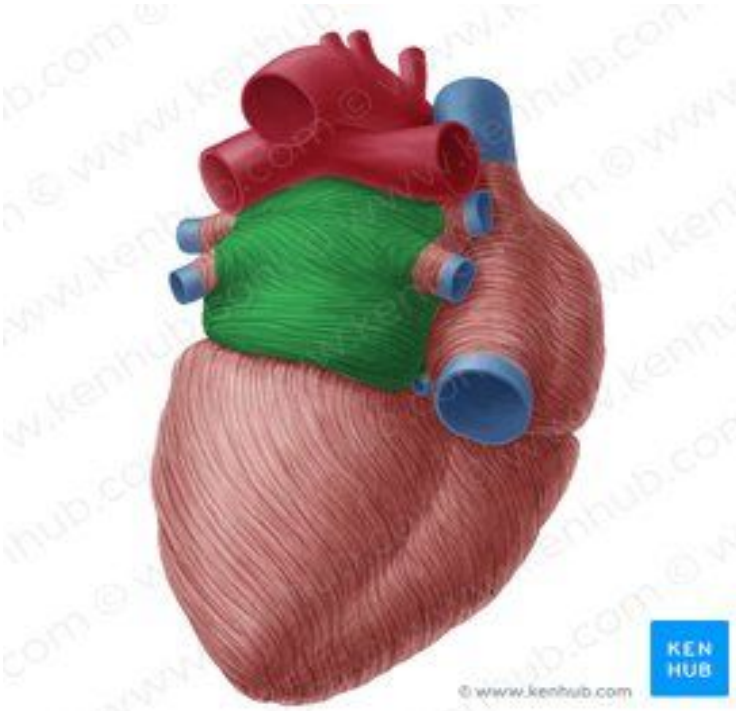
The Interventricular septum:

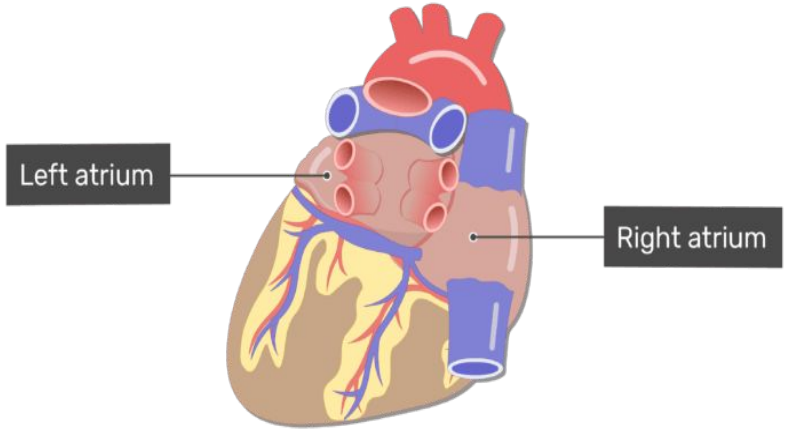
- The **interventricular septum** is indicated superficially by ***anterior and posterior interventricular grooves.***
- Its upper part is thin, membranous whereas lower part is thick muscular separating the two ventricles.



The Left Atrium:

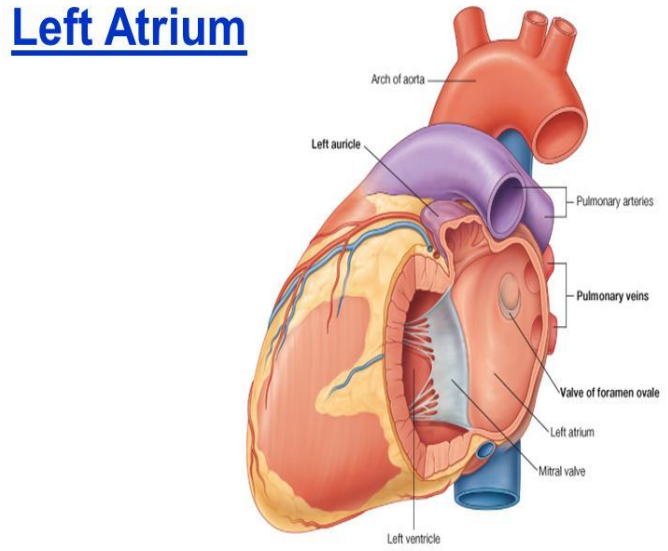
- It is a quadrangular chamber located posteriorly.
- Its appendage, the left auricle anteriorly overlaps the infundibulum of right ventricle.
- The left atrium receives oxygenated blood from lungs and pumps it to the left ventricle through left atrioventricular/bicuspid/mitral orifice.
- The mitral opening is gaurded by mitral/bicuspid valve.





Interior of the Left Atrium:

- Most of the internal part of left atrium is smooth-walled and develops from absorbed pulmonary veins which open into it.
- Whereas, rough part contains **pectinate muscles (musculi pectinati)** is present only in left auricle and it develops from primitive atrium.
- The **fossa lunata** is present in the inter-atrial septum.

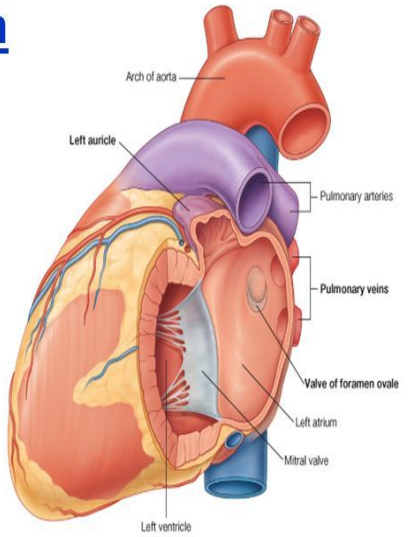


❖ Its appendage left auricle projects anteriorly to overlap the infundibulum of the right ventricle.

Tributaries/orifices of Left Atrium:

- Openings of **4 pulmonary veins, &**
- **Venae cordis minimi/thebesian veins**

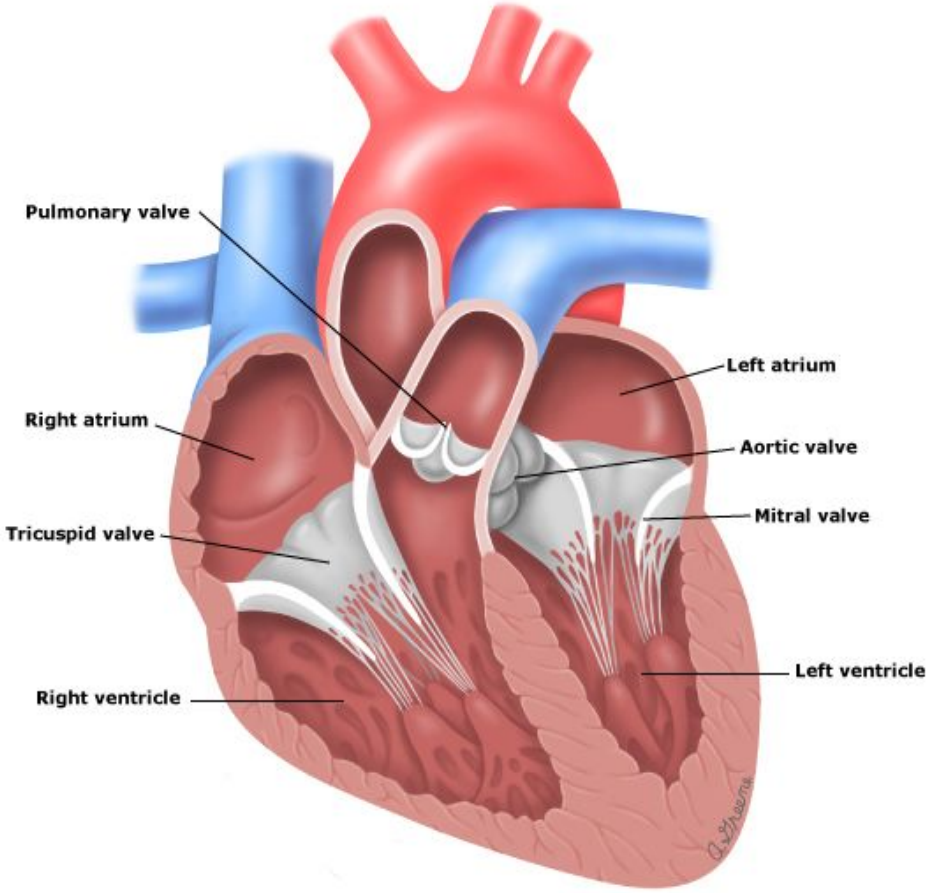
Left Atrium

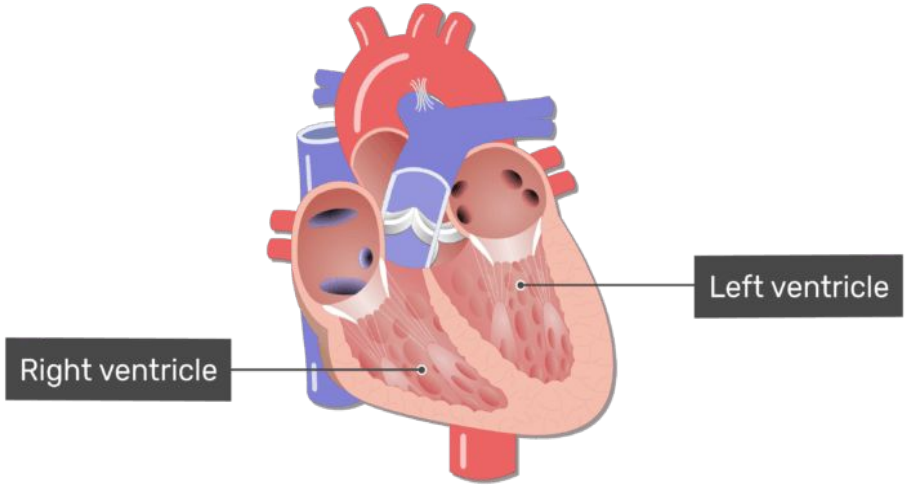


❖ Its appendage left auricle projects anteriorly to overlap the infundibulum of the right ventricle.

The Left Ventricle:

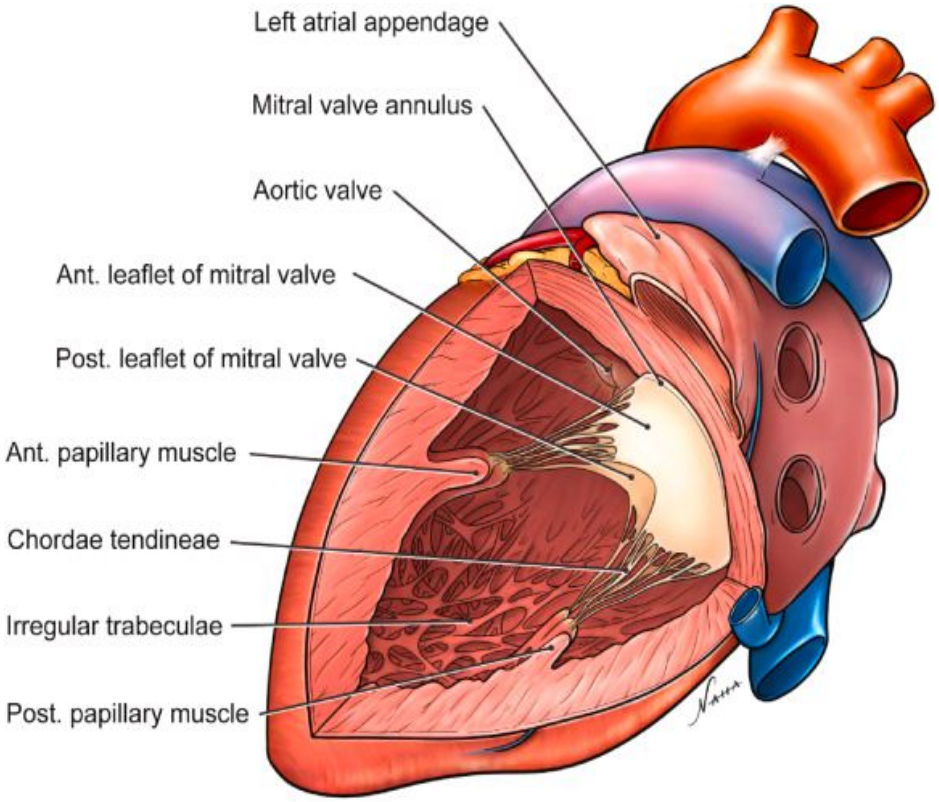
- The left ventricle receives oxygenated blood from left atrium and pumps it into the aorta.





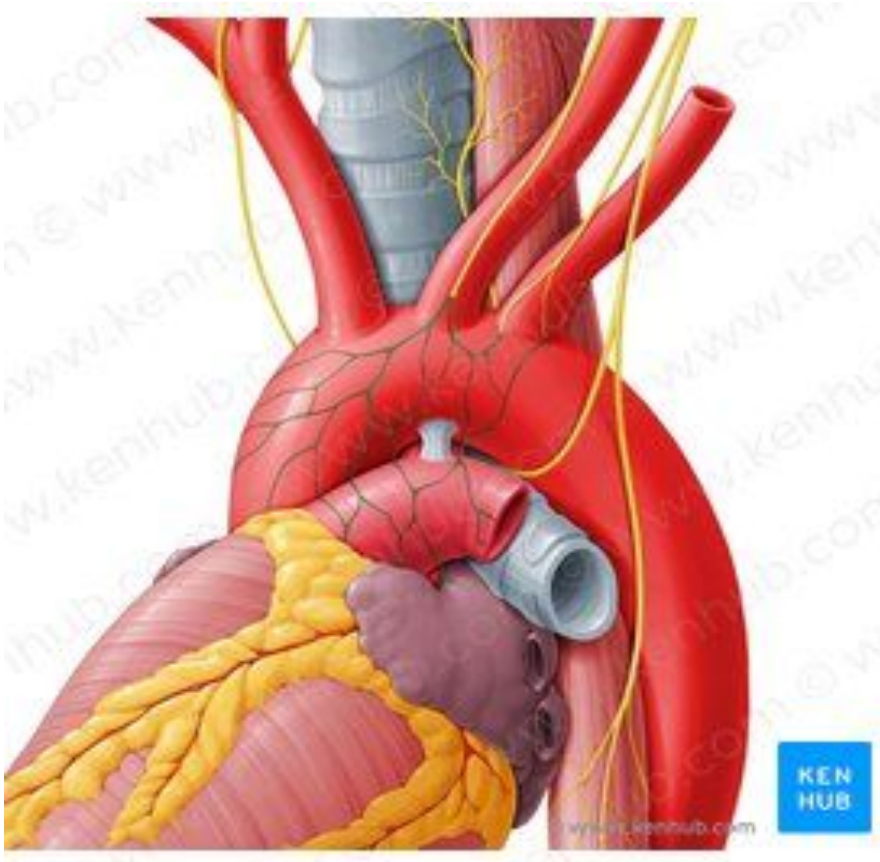
Interior of Left Ventricle:

- Interior of left ventricle has a lower **rough part** with **trabeculae carnae** and develops from primitive ventricle of heart tube, &
- An **upper smooth part**, the **aortic vestibule** which develops from bulbus cordis.
- The left ventricle has two orifices, the **mitral/left atrioventricular orifice** gaurded by mitral valve, & the **aortic orifice** gaurded by aortic valve.



Nerve Supply Of Heart:

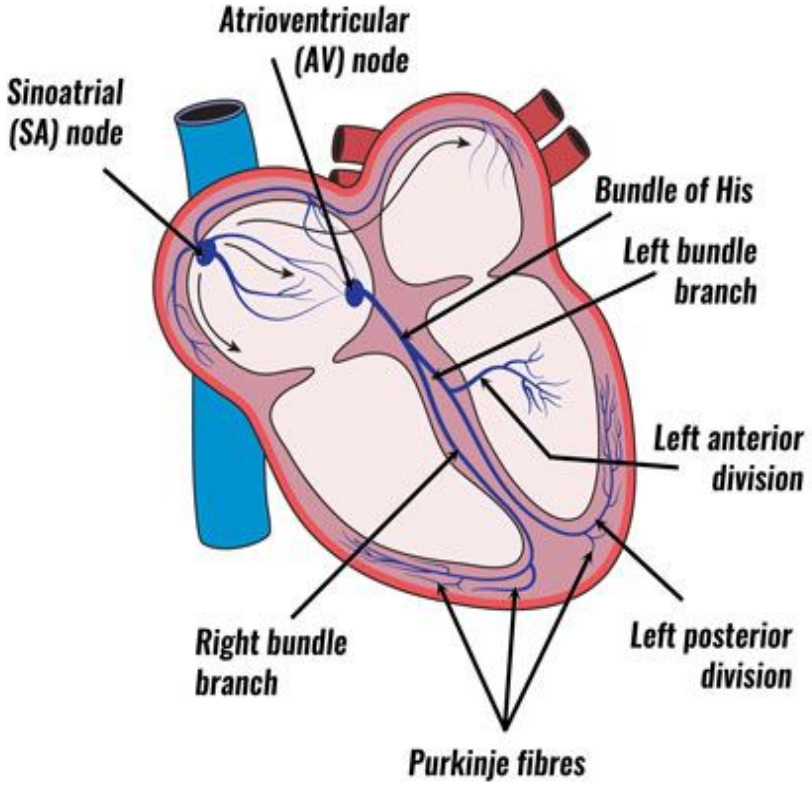
- **Parasympathetic:** fibers reach via *Vagus nerve*, cardio-inhibitory, i.e. upon stimulation, slows down the heart rate.
- **Sympathetic:** fibers are derived from *upper 4-5 thoracic segments* of spinal cord. They are cardio-accelaratory, i.e. upon stimulation increase the heart rate and dilate coronary arteries.
- The parasympathetic and sympathetic nerves form superficial and deep cardiac plexuses.



Conduction system of the Heart:

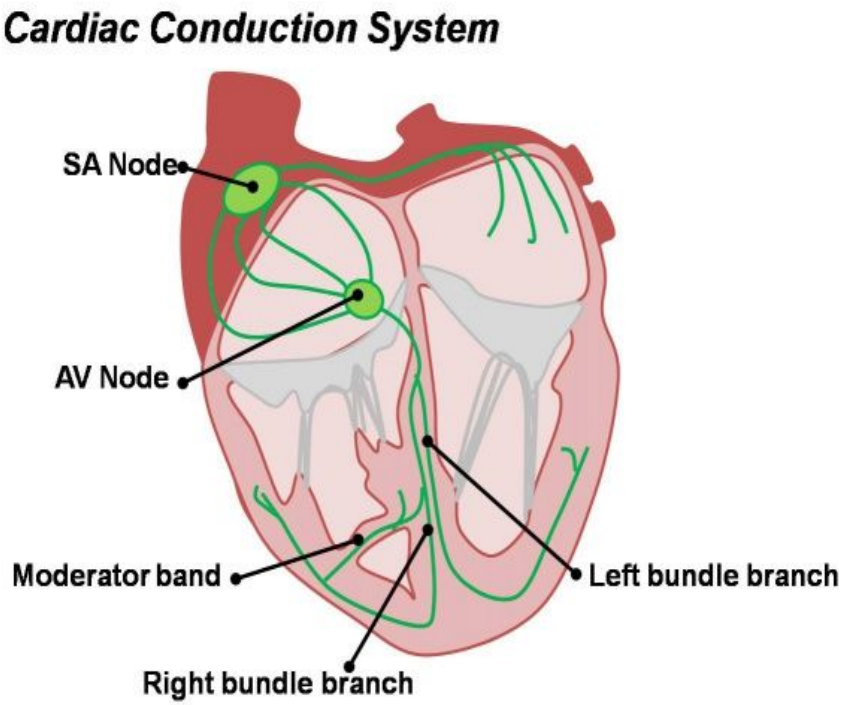
1. Sinoatrial/SA node:

- Also known as '***pacemaker'*** of the heart.
- Initiates the heart-beat.
- Situated at the atriocaval junction in the upper part of sulcus terminalis.
- Generates impulses at the rate of 70-100 beats/min.



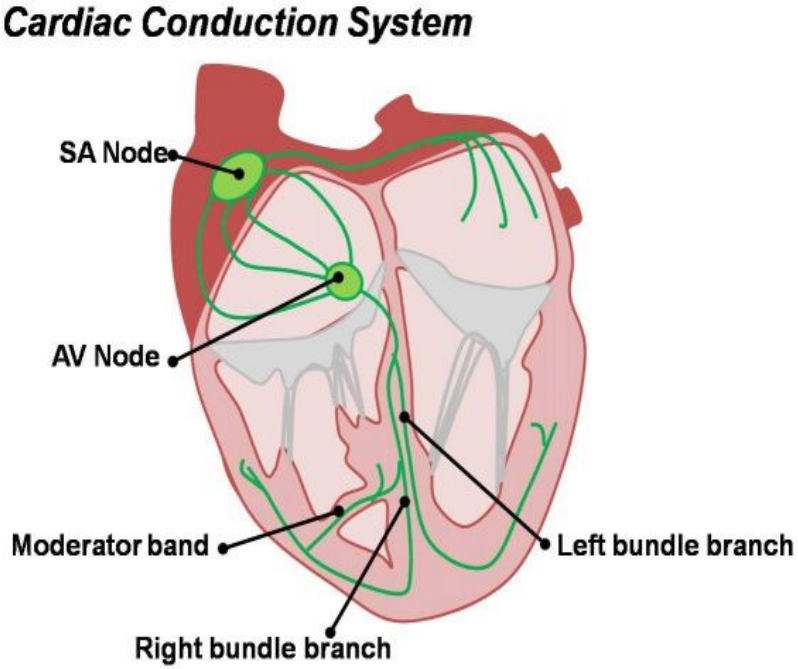
2. Atrioventricular/AV node:

- Situated on in the lower & posterior part of atrial septum above the opening of coronary sinus.
- Generates impulses at the rate of 40-60 beats/min.
- Smaller than the SA node.



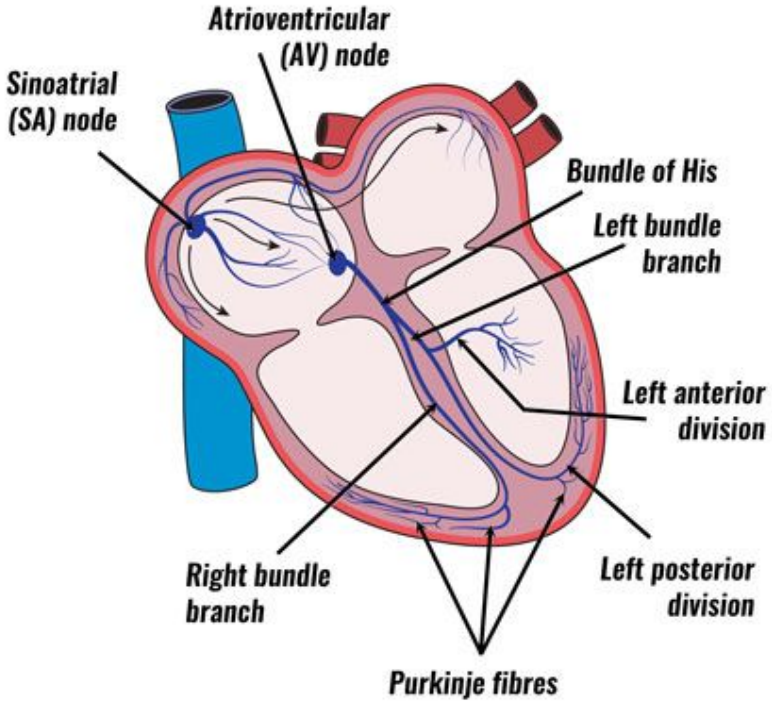
3. Atrioventricular bundle/bundle of His:

- The only muscular connection between atria & ventricles.
- Begins as the AV node & crosses the AV ring.
- Descends along the membranous part of interventricular septum.
- At the upper border of muscular part of septum, divides into right & left branches.



4. Right branch of AV bundle:

- Passes down the right side of interventricular septum.
- A large part enters the moderator band.
- Reaches the anterior wall of right ventricle & divides into Purkinje fibers.

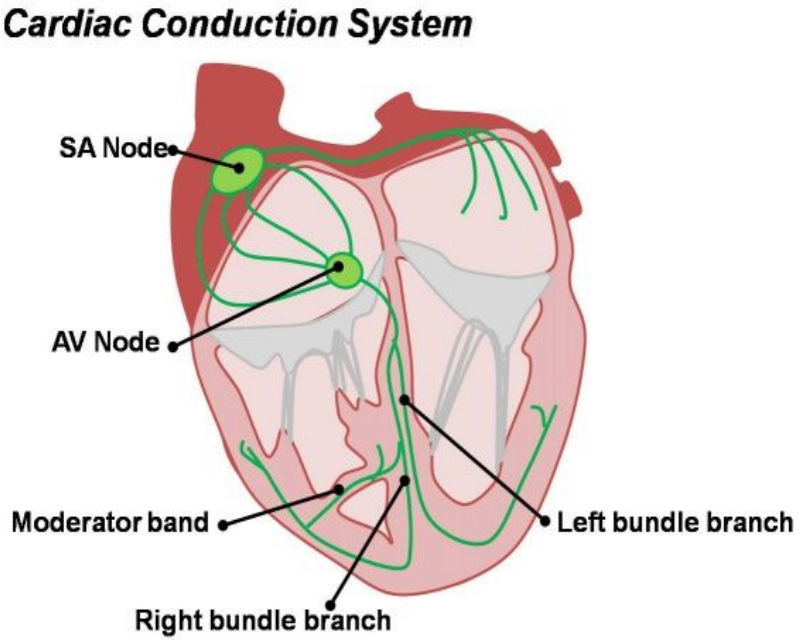


5. left branch of AV bundle:

- Descends on the left side of interventricular septum.
- Distributed to left ventricle after dividing into Purkinje fibers.

6. Purkinje fibers:

- Form subendocardial plexus.
- Usually possess double nuclei.
- Generate impulses at the rate of 20-35 beats/min.



References:

- **BD CHAURASIA'S HUMAN ANATOMY (Upper limb & Thorax)-Vol. 1**

6th edition

Chapter 18

Pages # 249-267


thank
you!